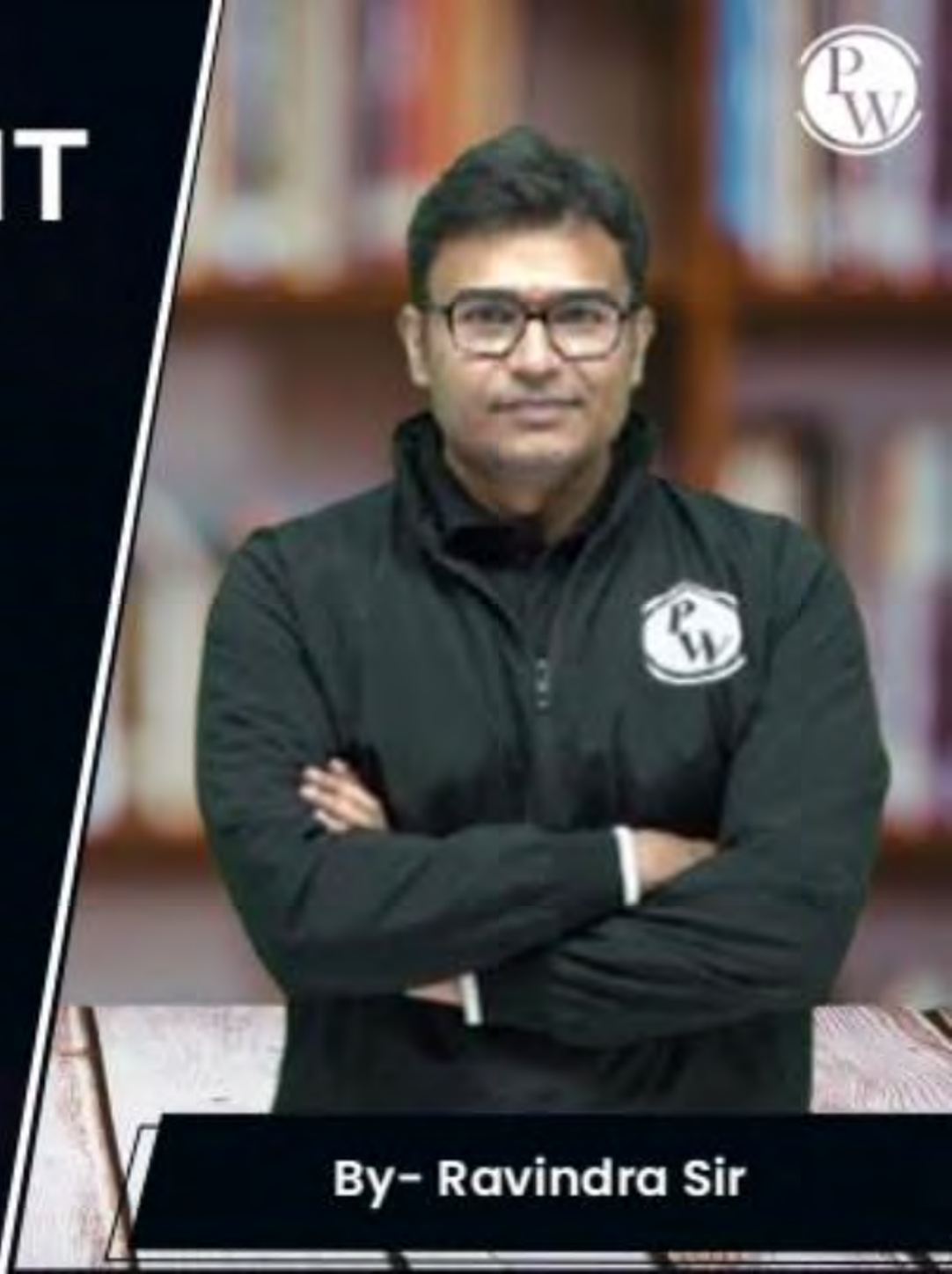
# Computer Science & IT

## COMPILER DESIGN
## (CD)

Parsers

Lecture No. 10



By- Ravindra Sir

# Recap of Previous Lecture



- Topic
- Topic
- Topic

Pauses

# Topics to be Covered



- Topic
- Topic
- Topic

Paneer

# Join Telegram

@RBRSIRPW





# Extra Ordinary Individuals: Stories to Ignite Student Motivation

***Shavarsh Karapetyan – The Swimmer Who Dove into Hell***

***Background:*** Armenian national swimming champion in the Soviet Union.

***Struggles:*** Saw a trolleybus sink into a freezing lake with 90 passengers. No gear, no hesitation.

***Achievements:*** Saved 20+ lives, repeatedly diving into glass, metal, and freezing water. Got pneumonia and severe injuries — ended his career.

***Impact:*** He traded medals for lives — and never looked back. If he could dive into shards for strangers, you can dive into tough chapters for your future.



# Extra Ordinary Individuals: Stories to Ignite Student Motivation

***B. Muthuraman***

***Background:*** Came from a modest Tamil Nadu family with limited means; no industrial lineage.

***Education:*** BTech in Metallurgical Engineering from IIT Madras, followed by MBA at XLRI Jamshedpur and executive training in Europe.

***Career Achievements:*** Joined Tata Steel in 1966; rose to Vice Chairman of Tata Steel and Chairman, Tata International.

***Impact:*** Key driver of Tata Steel's modernization; played critical strategic leadership roles in one of India's industrial pillars.



# Extra Ordinary Individuals: Stories to Ignite Student Motivation

***Mariappan Thangavelu – The Man Who Jumped Over Pity***

***Background:*** Born into poverty in Tamil Nadu. Crushed by a bus at age 5 — left permanently disabled.

***Struggles:*** Grew up with no privileges, no excuses. Trained barefoot. Mocked. Ignored.

***Achievements:*** Won Paralympic gold in high jump (2016), the first Indian ever in that category.

***Impact:*** He didn't jump for a medal. He jumped out of limits.

If he could rise above ridicule, you can rise above regret — one mock test at a time.



# Extra Ordinary Individuals: Stories to Ignite Student Motivation

***R. Gopalakrishnan***

***Background:*** Middle-class upbringing in Kolkata; no prominent business background.

***Education:*** Electronics degree from IIT Kharagpur, followed by Advanced Management Programme at Harvard Business School.

***Career Achievements:*** Held leadership roles at Hindustan Lever and Tata Sons; served on multiple board of directors including Tata Power, Castrol, and ABP.

***Impact:*** Brought strategic business excellence into Indian corporate core; helped scale consumer staples and industrial leadership.

$P \to SR/S \to ①$

$xR \to bSR/bS \to ②$

$S \to wbS/w$

$w \to Lyw/L$

$L \to id$

$P \to SbSR/SbS/S$

$P \to SbP/SbS/S$

$S \to wbS/w$

$w \to L*w/L$

$L \to id$

$id \doteq> *$ $\quad b \lessdot b$

$* \lessdot *$

$* \doteq> b$

$b \lessdot *$

Precedence relation table

| | id | * | b | $ |
|---|---|---|---|---|
| id | — | $\doteq>$ | $\doteq>$ | $\doteq>$ |
| * | $\lessdot$ | $\lessdot$ | $\doteq>$ | $\doteq>$ |
| b | $\lessdot$ | $\lessdot$ | $\lessdot$ | $\doteq>$ |
| $ | $\lessdot$ | $\lessdot$ | $\lessdot$ | — |

The disadvantage of precedence relation table is that the size is $O(n^2)$

So we go for precedence function table

## Procedure for Converting precedence relation table to precedence function table :-

Input is precedence relation table

Step1 : For every terminal a Create $f_a$ and $g_a$

Step2 :- Divide the symbols into groups by using $\doteq$ relation ($\doteq$)

Ex :- $a \doteq b$ then $f_a$ and $g_b$ are combined into single group

Step3 :- Create a digraph with groups in step 2 as nodes and edges given by $<\cdot$ or $\cdot>$

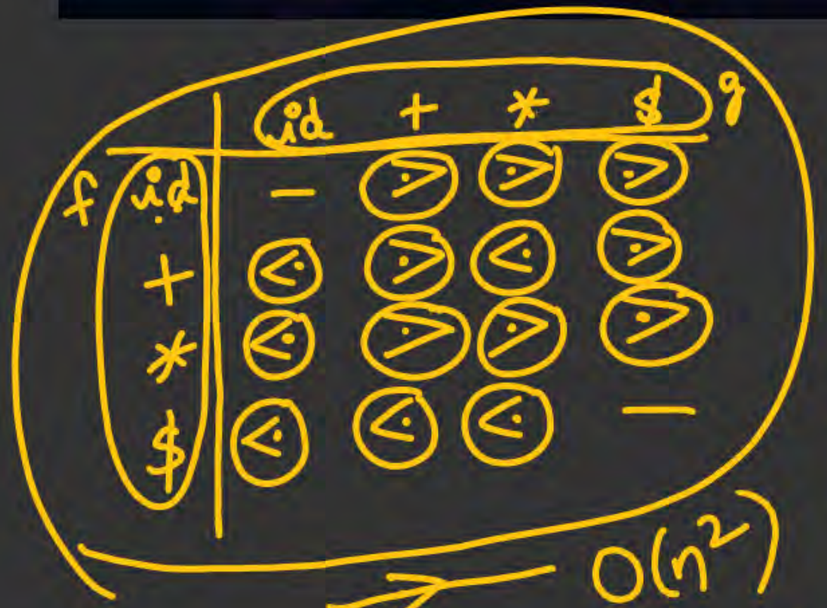
For example if $a > b$, then add edge as $f_a \rightarrow g_b$

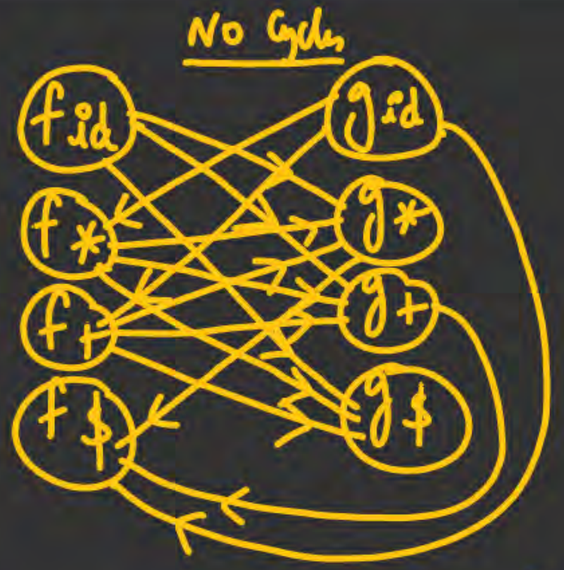
For example if $a < b$, then add edge as $f_a \leftarrow g_b$

Add an edge for each $\leq$ or $\geq$ relation in the table

Step 4:- Check the digraph for cycles. If there are any cycles stop the procedure and conclude that the table cannot be converted to a function table

Step 5 :- If there are no cycles, that is the digraph is acyclic then length of the longest path from $f_a$ gives $f(a)$ for each terminal 'a'




f(id) = 4 ✓ g(+) = 1 ✓

$f_{id} \rightarrow g_* \rightarrow f_+ \rightarrow g_+ \rightarrow f_\$$

$g_{id} \rightarrow f_* \rightarrow g_* \rightarrow f_+ \rightarrow g_+ \rightarrow f_\$$

O(2n)

| | id | + | * | $ |
|---|---|---|---|---|
| f | 4 | 2 | 4 | 0 |
| g | 5 | 1 | 3 | 0 |

| | a | ( | ) | , | $ |
|---|---|---|---|---|---|
| a | ○ | ○ | > | > | > |
| ( | < | < | = | < | ○ |
| ) | ○ | ○ | > | > | > |
| , | < | < | > | > | ○ |
| $ | < | < | ○ | ○ | ○ |

| | a | ( | ) | , | $ |
|---|---|---|---|---|---|
| f | 2 | 0 | 2 | 2 | 0 |
| g | 3 | 3 | 0 | 1 | 0 |

The disadvantage of function table is that it doesn't have end entries. So we have to maintain a seperate table for end entries and see it before seeing function table

# THANK - YOU